Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

Rows: 3,900

Columns: 18

Key Features:

Customer demographics (ID, Age, Gender, Location, Subscription Status)

Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)

Shopping behavior (Discount Applied, Subscription Status, Previous Purchases, Frequency of Purchases, Review Rating, Payment Method)

Missing Data : The data was there and good to go

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

Data Loading: Imported the dataset using pandas.

Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

```
df.describe(include='all')
```

Python

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Discount Applied	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3900.000000	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.749949	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716223	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.700000	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	50.000000	NaN	NaN

Missing Data Handling: Checked for null values and missing values using the median rating of each product category.

Column Standardization: Renamed columns to snake case for better readability and documentation.

Feature Engineering: Created (age_group) column by binning customer ages; created (purchase_frequency_days)column from purchase data.

Database Integration: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

I performed structured analysis in PostgreSQL to answer key business questions:

. Revenue by Gender – Compared total revenue generated by male vs. female customers.
Insight: Male customers generated $157,890 (about 68% of total revenue) vs. $75,191 for female customers, roughly 2x more. This largely reflects the customer base, since about 68% of customers are male.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

. High-Spending Discount Users – Identified customers who used a discount but still spent above the average purchase amount.
Insight: 839 customers used a discount and still spent more than the average, so discounts do not always mean lower-value purchases.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91

. Top 5 Products by Rating – Found the products with the highest average review ratings.
Insight: Gloves (3.86), Sandals (3.84), Boots (3.81), Hat (3.81), and Skirt (3.78) lead. The gap between them is small, so ratings are consistently good rather than driven by one standout.
Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.81
4	Hat	3.81
5	T-shirt	3.78

. Which age group contributes the most according to revenue? – Calculated total revenue for each age group and ranked them from highest to lowest
Insight: Young Adults contribute the most with $62,143, followed by Middle-aged ($59,197), Adult ($55,978), and Senior ($55,763).

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle Aged	59197
3	Adult	55978
4	Senior	55763

. Which payment method was used the most, and how much total revenue did it generate?
Compared payment methods by number of purchases and total revenue.
Insight: credit card generated the most in revenue with $40,310 and was used 671 time , but paypal was used the most with total of 677 time and generated $40,109, followed by cash ($40,002) and used (670), debit card ($38,742) and used (636), venmo ($37,374) and used (634) and last was bank transfer generated the least in revenue ($36,544) and was used the least (612).

	payment_method text	total_purchases bigint	total_revenue numeric
1	PayPal	677	40109
2	Credit Card	671	40310
3	Cash	670	40002
4	Debit Card	636	38742
5	Venmo	634	37374
6	Bank Transfer	612	36544

. How do different subscription statuses compare in terms of customer count, average spend, and total revenue?

Compared subscribers vs. non-subscribers by customer count, average spend, and total revenue.

Insight: Only 1,053 customers (27%) are subscribers, generating $62,645. The 2,847 non-subscribers generated $170,436. Average spend is almost identical ($59.49 vs. $59.87), so subscribing does not yet increase spending.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

. Which items relied most on discounts to drive sales, and how many times were they purchased?

Calculated each item's discount rate (percentage of purchases made with a discount) alongside its total purchases, and listed the top 5.

Insight: Hat have been sold (154) times and (50.00%) of the sales was with the discount applied, Sneakers was (145) and(49.66%), Coat (161) and (49.07%), Sweater (164) and (48.17%), and Pants (171) and (47.37%). Hats relied most heavily on discounts, with 50% of their purchases made using a discount.

	item_purchased text	total_purchases bigint	discount_rate numeric
1	Hat	154	50.00
2	Sneakers	145	49.66
3	Coat	161	49.07
4	Sweater	164	48.17
5	Pants	171	47.37

. For each purchase frequency level, how many customers are there and how many discounts were applied?

Grouped customers by purchase frequency, counting the customers in each level and the discounts applied to them.

Insight: The customer base is balanced across all 7 frequency levels, with each holding about 14-15% of customers. Discount usage follows the same pattern, with every level accounting for roughly 13-15% of all 1,677 discounts, so no single frequency group drives discount usage.

	frequency_of_purchases text	total_customers bigint	total_discounts bigint
1	Weekly	539	234
2	Quarterly	563	241
3	Monthly	553	237
4	Fortnightly	542	243
5	Every 3 Months	584	258
6	Bi-Weekly	547	226
7	Annually	572	238

. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

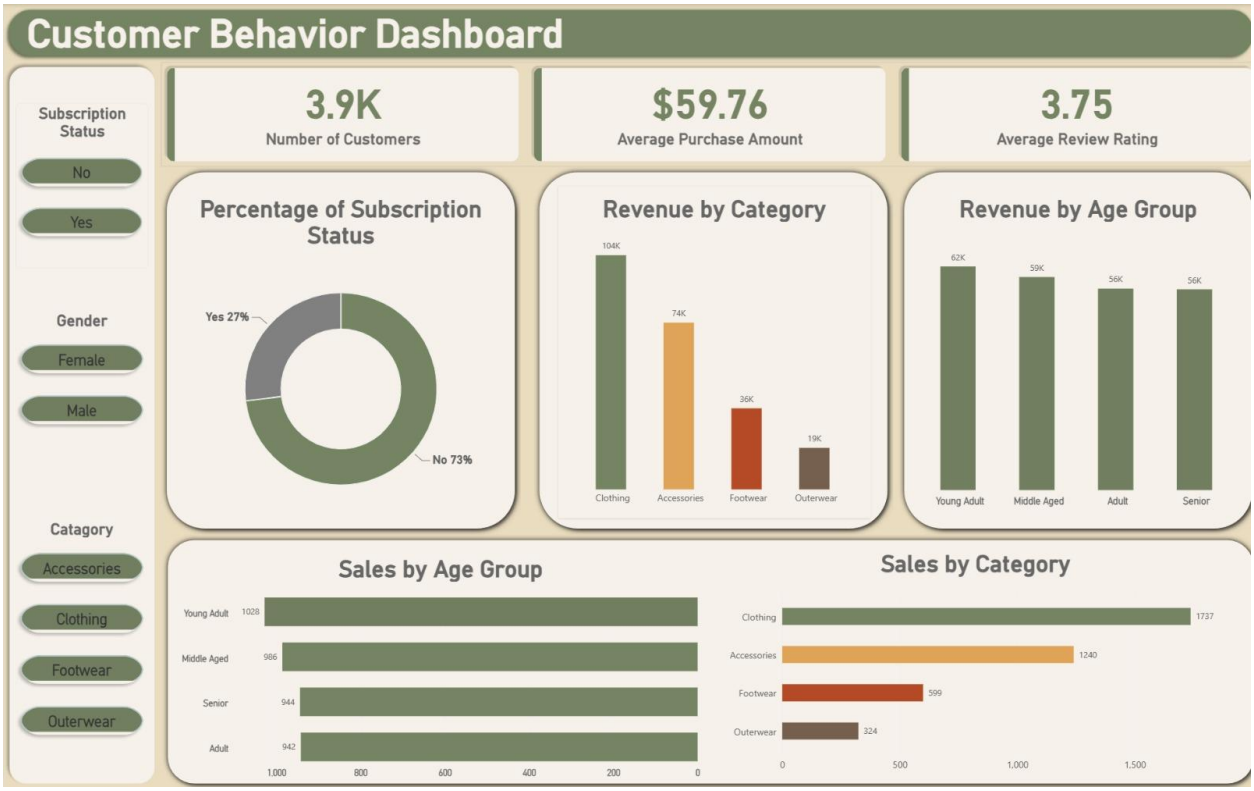
Counted customers with more than 5 previous purchases by subscription status.

Insight: Of 3,476 repeat buyers, 958 (about 27.6%) are subscribers, Repeat buyers are somewhat more likely to subscribe, but 72% still have not, so there is room to convert them.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

Boost Subscriptions – Only 27% of customers subscribe, and subscribers spend about the same as non-subscribers ($59.49 vs. $59.87). Add exclusive benefits such as early access, free shipping, or members-only pricing, so that subscribing gives customers a reason to spend more.

Convert Repeat Buyers into Subscribers – Repeat buyers subscribe only slightly more often (about 27.6%) than other customers, yet 72% of them are still not subscribed. They already know the brand, which makes them the easiest group to target with subscription offers.

Review Discount Policy – Discounts are applied to about 43% of purchases, and they are spread evenly across all purchase frequencies, so they are not rewarding any particular behavior. Since 839 customers spent above the average even with a discount, test whether some of them would have bought at full price. Reduce discounts on items that rely on them heavily (Hat, Sneakers, Coat, Sweater, Pants, all near 50%) or limit them to slower-moving items.

Target Discounts Strategically – Instead of broad discounts, offer them to occasional buyers (Annually, Every 3 Months) to encourage them to buy more often, and use non-price rewards for frequent buyers.

Product Positioning – Feature top-rated products (Gloves, Sandals, Boots, Hat, Skirt) in campaigns and use their ratings as social proof. The rating gaps are small, so highlight these items together as a group rather than picking one.

Targeted Marketing by Age Group – Young Adults generate the most revenue ($62,143), but all age groups are within about 3 percentage points of each other. Run campaigns tailored to Young Adults while keeping broad appeal for the other groups, as no age group can be ignored.

Engage Female Customers – Male customers generate about twice the revenue of female customers, largely because they make up most of the customer base. Create campaigns and product ranges that attract more female shoppers, which is the largest growth opportunity in the data.